

Calcurion Maths Worksheets

A-Level Mathematics

Topic: Surds & Indices

Medium & Hard

Section A – Medium

1. Simplify fully

$$\sqrt{128} - 2\sqrt{32} + \sqrt{18}.$$

[4]

2. Express $\sqrt{432}$ in the form $a\sqrt{b}$, where a and b are positive integers and b is square-free. [3]

3. Rationalise the denominator of

$$\frac{6}{2\sqrt{5} - \sqrt{3}}.$$

[4]

4. Simplify

$$\frac{(2\sqrt{5})^3}{\sqrt{20}}.$$

[5]

5. Express $\frac{a^{-5}b^2}{c^{-3}}$ using only positive indices. [3]

6. Simplify fully

$$\left(\frac{32x^{10}}{243y^{15}}\right)^{\frac{2}{5}}.$$

[5]

7. Solve the equation

$$4^{x+\frac{1}{2}} = 8^{x-1}.$$

[4]

8. Simplify

$$\frac{m^5n^{-4}}{m^{-3}n^2} \times (mn)^{-2}.$$

[5]

Section B – Hard

9. Expand and simplify

$$(3\sqrt{2} - \sqrt{5})(3\sqrt{2} + \sqrt{5}).$$

Hence, or otherwise, express

$$\frac{\sqrt{2} + 1}{3\sqrt{2} - \sqrt{5}}$$

in the form $a\sqrt{2} + b\sqrt{5} + c\sqrt{10}$, where a , b and c are rational.

[7]

10. Solve the equation

$$\sqrt{5x + 1} - \sqrt{2x - 1} = 2$$

for $x \geq \frac{1}{2}$.

[8]

11. Express

$$\frac{5 + \sqrt{7}}{\sqrt{28} - 3}$$

in the form $a + b\sqrt{7}$, where a and b are rational numbers.

[6]

12. Solve for x

$$9^{x+1} \cdot 3^{2x-1} = 27^x.$$

[6]

13. Given that

$$\left(\frac{x^9}{64y^{-12}}\right)^{-\frac{1}{3}} = \frac{4y^k}{x^3},$$

find the value of the constant k .

[6]

14. Simplify fully

$$\frac{\sqrt{75} + \sqrt{48}}{\sqrt{3}} - \frac{\sqrt{98}}{\sqrt{2}} + \sqrt{32}.$$

[6]

15. Solve the equation

$$4^x - 5 \cdot 2^x + 4 = 0.$$

[7]

End of Worksheet • Show all working clearly in the spaces provided.